

Branch Prediction

static

Dynamic

1-Bit Predictor

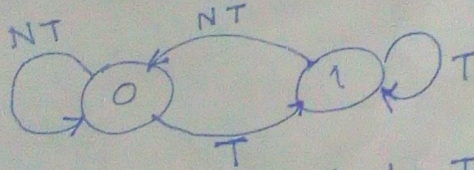
2-Bit Predictor

1-Bit Predictor:

1-bit $\Rightarrow$ 2 States

state 0 $\Rightarrow$ Branch will be Not Taken

state 1 $\Rightarrow$ Branch will be Taken



Prediction

Assume branch is predicted to be Taken for a 'for' loop
for (i=0; i<5; i++)

iterations	0,	1,	2,	3,	4,	5
Pred.	T	T	T	T	T	T
Reality	✓	✓	✓	✓	✓	X
States	①	①	①	①	①	①
	①	①	①	①	①	①

This means if the
Next time occurs branch
same loop will not be taken
which will be wrong
So, 1-bit predictor makes
two mistakes
one while exiting
& second while
entering again